

SHUTIAN LUO

CONTACT

Postdoctoral Researcher
Department of Computer Science, University of Virginia
Location: Charlottesville, VA, U.S.

Phone: +1 (475) 2247063
Email: ksy8xs@virginia.edu
[Homepage](#) and [Google Scholar](#)

EDUCATION

University of Chinese Academy of Sciences Sept. 2016 - Jan. 2023

PhD in Computer Application Technology

Dissertation Advisor: Prof. [Chengzhong Xu](#). Thesis: Research on Key Technologies of Intelligent Resource Management for Cloud Native Microservices Workload.

Shenzhen University Sept. 2012 - Jul. 2016

B.Eng in Electronics Engineering

Postgraduate Recommendation (**3 out of 283**)

RESEARCH INTERESTS

- AI infrastructure and systems for machine learning, with a focus on large-scale LLM and MoE serving on heterogeneous CPU-GPU platforms (e.g., NVIDIA Superchips).
- Cloud-native and GPU-centric runtimes and resource management: microservices, serverless platforms, and RDMA/DSM-based systems for large-scale, latency-sensitive AI workloads.

WORK EXPERIENCE

Postdoc in University of Virginia Feb. 2025 - Present

Advisor: Prof. [Haiying Shen](#)

- Designed zero-copy KV-cache offloading and asymmetric computation CUDA kernels on NVIDIA Superchips with a unified CPU-GPU memory hierarchy, improving GPU memory efficiency and latency for long-context LLM inference.
- Designed an on-demand expert streaming system on NVIDIA Superchips that replaces MoE token all-to-all communication with expert streaming, significantly improving inference throughput.

Postdoc in Yale University Jun. 2023 - Jan. 2025

Advisor: Prof. [Lin Zhong](#) and Co-advisor: Prof. [Anurag Khandelwal](#)

- Explored the trade-off between performance and programming transparency for data-parallel workloads in RDMA-based distributed shared memory (DSM) systems.

Research Assistant in University of Macau Apr. 2021 - May 2023

[Cloud and Distributed Systems Lab](#)

- Investigated workload variability in microservices using deep learning for predictive autoscaling, and proposed a resource management framework to ensure SLA requirement in shared environments.

Research Intern in Alibaba Cloud Nov. 2018 - Jan. 2023

- Designed resource management system for online shopping application in Alibaba to handle extremely high load during Double 11 Festival (similar to Black Friday) for high performance.

PUBLICATIONS

1. [ASPLOS'25] Shutian Luo, Jianxiong Liao, Huanle Xu, Zhi Zhou and Chengzhong Xu. Embracing Imbalance: Dynamic Load Shifting among Microservice Containers in Shared Clusters. In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems. 2025.
2. [SoCC'25] Yanying Lin, Shuaipeng Wu, Shutian Luo, and Hong Xu, Haiying Shen, Chong Ma, Min Shen, Le Chen, Chengzhong Xu, Lin Qu, Kejiang Ye. Understanding Diffusion Model Serving in Production: A Top-Down Analysis of Workload, Scheduling, and Resource Efficiency. In Proceedings of the 16th ACM Symposium on Cloud Computing. 2025.
3. [HPCA'25] Liao Chen, Chenyu Lin, Shutian Luo, Huanle Xu and Chengzhong Xu. Grad: Intelligent Microservice Scaling by Harnessing Resource Fungibility. In Proceedings of the 31st IEEE International Symposium on High-Performance Computer Architecture. 2025.
4. [ISCA'24] Liao Chen*, Shutian Luo*, Chenyu Lin, Zizhao Mo, Huanle Xu, Kejiang Ye and Chengzhong Xu. Derm: SLA-aware Resource Management for Highly Dynamic Microservices. In Proceedings of the 51th International Symposium on Computer Architecture. 2024.
5. [ICDCS'24] Yanying Lin, Yanbo Li, Yingfei Tang, Shutian Luo, Haiying Shen, Chengzhong Xu and Kejiang Ye. Quart: Latency-Aware FaaS System for Pipelining Large Model Inference. In Proceedings of the 44th IEEE International Conference on Distributed Computing Systems. 2024.
6. [ASPLOS'23] Shutian Luo, Huanle Xu, Kejiang Ye, Guoyao Xu, Liping Zhang, Jian He, Guodong Yang, and Chengzhong Xu. Erms: Efficient Resource Management for Shared Microservices with SLA Guarantees. In Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems. 2023.
7. [ACM SoCC'22] Shutian Luo, Huanle Xu, Kejiang Ye, Guoyao Xu, Liping Zhang, Jian He, and Chengzhong Xu. The Power of Prediction: Microservice Auto Scaling via Workload Learning. In Proceedings of the 13th ACM Symposium on Cloud Computing. 2022.
8. [ACM SoCC'21] Shutian Luo, Huanle Xu, Chengzhi Lu, Kejiang Ye, Guoyao Xu, Liping Zhang, Yu Ding, Jian He, and Chengzhong Xu. Characterizing Microservice Dependency and Performance: Alibaba Trace Analysis. In Proceedings of the 12th ACM Symposium on Cloud Computing. 2021. **(Best Paper Award.)**
9. [IEEE TPDS] Shutian Luo, Huanle Xu, Chengzhi Lu, Kejiang Ye, Guoyao Xu, Liping Zhang, Jian He, and Chengzhong Xu. An In-depth Study of Microservice Call Graph and Runtime Performance. In Proceedings of IEEE Transactions on Parallel and Distributed Systems. 2022.
10. [ACM TOCS] Shutian Luo, Chenyu Lin, Kejiang Ye, Guoyao Xu, Liping Zhang, Guodong Yang, Huanle Xu and Chengzhong Xu. Optimizing resource management for shared microservices: a scalable system design. In Proceedings of ACM Transactions on Computer Systems. 2024.
11. [ISPA'24] Chenyu Lin, Shutian Luo, Huanle Xu. Exploring Imbalances among Microservice Containers in Large Cloud Platforms. In Proceedings of IEEE ISPA. 2024.

WORKING PAPERS

1. Shutian Luo and Haiying Shen. No Buffer, No Bottleneck: Efficient Zero-Copy KV Cache Offloading for Long-Context LLMs.
2. Shutian Luo and Haiying Shen. C2CMoE: On-Demand Expert Streaming over NVLink-C2C for Mixture-of-Experts Inference.
3. Shutian Luo, Chenxi Dai, Chenyu Lin, Liao Chen, Huanle Xu, Pan Zhou, Haiying Shen and ChengZhong Xu. Nirem: A Non-intrusive Resource Management System for Microservices in Public Clouds.
4. Shuo Huang, Shutian Luo, Lin Zhong and Mingsung Kim. Pagoda: A Real-Time and End-to-End Large MIMO vRAN System.
5. Liao Chen, Chenyu Lin, Junlin Chen, Shutian Luo, Huanle Xu and ChengZhong Xu. Cremes: Cost-Efficient and Reliable Microservice Execution on Spot Instances.

WORKING PROCESS

1. [Shutian Luo](#), Rui Yang, Caihua Li, Yue Cheng, Wei Wang and Haiying Shen. MIGServe: Serverless LLM Serving on MIG-Partitioned GPUs with Token-Level Scheduling.

HONORS/ AWARDS

- ACM SoCC'2021 **Best Paper** Award.
- 2024 China Computer Federation (CCF) Computer Architecture Outstanding Ph.D (**5 awarded annually**).
- 2024 IEEE Technical Community on Scalable Computing (TCSC) Outstanding Ph.D.
- 2024 Chinese Academy of Sciences Outstanding Ph.D. (**the first Ph.D.** to receive this award in the past 16 years at Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences).
- **Alibaba Star offer** (P7 level, highly professional expert). Alibaba Star is the highest offer for Ph.D graduates, with no more than 10 positions offered annually at Alibaba.
- Outstanding Science Research Intern at Alibaba Cloud in 2021 (**awarded to only 12 interns**).
- Dean's award for academic performance (top 1%) in 2021

OPEN-SOURCED PROJECTS

- Released two versions of Alibaba microservices traces: 100GB v2021[\[Github\]](#) and 2TB v2022[\[Github\]](#)
- Designed an efficient resource management System for microservices [\[Github\]](#), earning three ACM badges for availability, functionality, and reproducibility from the conference committee.

TEACHING & MENTORING

Guest Lecturer in Cloud-Native Computing and Microservices 2025
University of Virginia

- Designed and delivered guest lectures on cloud-native computing and microservices, including microservice dependency graphs, tail latency, and QoS-aware resource management using Alibaba Cloud traces.
- Prepared lecture slides and quizzes to help students reason about large-scale microservice workloads and resource management policies.

Teaching Support in Systems Courses 2021 – 2023
University of Macau

- Assisted with courses in operating systems and distributed systems, including answering questions on concurrency, scheduling, and fault tolerance.
- Helped design and review programming assignments and projects that connect core systems concepts to real-world cloud workloads.

Research Mentoring 2019 – Present

- Mentored research interns and junior Ph.D students on projects in microservice resource management and GPU/LLM infrastructure.
- Guided mentees through the full research cycle, including problem formulation, system design and implementation, experimental evaluation, and paper writing.

COMMUNITY SERVICES

- Reviewer for IEEE Transactions on Parallel and Distributed Systems, Journal of Cloud Computing, IEEE/ACM Transactions on Networking, IEEE Transactions on Mobile Computing, IEEE Transactions on Cloud Computing, Journal of Supercomputing
- TPC member for COINS 2025, SmartData-2024, HDIS 2023, IEEE ScalCom 2022.
- Shadow PC for EuroSys 2023

SKILLS

C++, Python, CUTLASS, CUDA, vLLM, NCCL, Kubernetes, RDMA, Pytorch, OpenMP, Spark

SELECTED TALKS & PRESENTATIONS

- Characterizing Microservice Dependency and Performance: Alibaba Trace Analysis. ACM SoCC'21. Conference presentation.
- The Power of Prediction: Microservice Auto Scaling via Workload Learning. ACM SoCC'22. Conference presentation.
- Erms: Efficient Resource Management for Shared Microservices with SLA Guarantees. ASP-LOS'23. Conference presentation.
- AI Infrastructure for Large-Scale LLM Serving on NVIDIA Superchips. Systems/AI seminar, University of Pennsylvania, 2025. Invited talk.